



BEGINNER-FRIENDLY SETUP MANUAL

Wind Turbine Design Lab

How to install, run, update, and troubleshoot the simulator on a Windows PC or Mac - no coding experience required.

What you need

Internet connection
About 10-20 minutes
Permission to install apps

What it costs

The project is free to download.
No cloud hosting or subscription is required.

Repository

<https://github.com/TBK0749/wind-turbine-design-lab>

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START HERE

Before you begin

This guide assumes you have never used Git, Python, or a command-line terminal.

What this installation does

The simulator will run only on your own computer. Git downloads the project from GitHub. uv prepares the correct Python environment and installs the required software packages. Streamlit opens the simulator in your web browser.

1. GitHub Stores the project online	2. Git Copies and updates it	3. uv Prepares Python	4. Browser Shows the simulator
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Choose your computer

Windows 10 or 11	Follow pages 3-5
macOS	Follow pages 6-7
Already installed?	Go to page 8 to run the app

Words used in this guide

O K	Terminal: a text window where you type instructions. On Windows, use PowerShell. On Mac, use Terminal.
O K	Command: one line of text you type or paste, then run by pressing Enter.
O K	Project folder: the folder named <code>wind-turbine-design-lab</code> .
O K	Local address: a private web address that works only while the app is running on your computer.

Important

You do not need a GitHub account to download this public project. You also do not need to click Deploy or enter payment information.

WINDOWS - PART 1

Install Git and uv

Complete these steps once. Future app updates will be much shorter.

1 Open PowerShell

Click **Start**, type **PowerShell**, then open **Windows PowerShell** or **Terminal**. You will type the commands below into this window.

2 Check whether Git is already installed

Type the following command and press Enter.

```
git --version
```

If you see a version number

Git is ready. Skip to Step 4.

If Windows says git is not recognized

Open git-scm.com/install/windows in your browser. Download the x64 installer for a normal Intel/AMD PC, run it, and accept the default options. Close and reopen PowerShell afterward.

3 Confirm Git

Run the check again. A version number means the installation worked.

```
git --version
```

4 Install uv

Copy the complete line below, paste it into PowerShell, and press Enter.

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

5 Restart PowerShell and confirm uv

Close PowerShell completely, open it again, then run:

```
uv --version
```

Why there is no separate Python installer

uv can automatically download the Python version required by this project. This avoids several common beginner setup problems.

WINDOWS - PART 2

Download and install the project

These commands create a project folder inside your Documents folder.

1 Go to Documents

Open PowerShell and run:

```
cd $HOME\Documents
```

2 Download the project from GitHub

Copy and run the following command:

```
git clone https://github.com/TBK0749/wind-turbine-design-lab.git
```

What you should see

PowerShell should display Cloning into 'wind-turbine-design-lab'... and then return to the prompt. This may take a moment.

3 Enter the project folder

Run:

```
cd wind-turbine-design-lab
```

4 Prepare Python and install the app packages

Run the command below. The first installation can take several minutes because files must be downloaded.

```
uv sync
```

Do not close the window too early

Wait until the progress finishes and PowerShell shows a new prompt. Messages about downloading, resolving, or installing packages are normal.

5 Start the simulator

Run:

```
uv run streamlit run app/main.py
```

First-run email question

If Streamlit asks for an email address, leave it blank and press Enter. You do not need to register.

WINDOWS - PART 3

Open it again next time

After the first installation, starting the simulator takes only three commands.

Everyday startup

1 Open PowerShell

Click Start and open PowerShell or Windows Terminal.

2 Go to the project folder

Run:

```
cd $HOME\Documents\wind-turbine-design-lab
```

3 Start the app

Run:

```
uv run streamlit run app/main.py
```

4 Open the web page

The browser normally opens automatically. If it does not, open Chrome, Edge, or Firefox and enter:

```
http://127.0.0.1:8501
```

How to stop the app

Return to the PowerShell window that is running Streamlit. Press **Ctrl + C** once. Closing only the browser tab does not stop the app.

Keep PowerShell open while using the simulator

The terminal is the engine running the local website. If you close it, the browser page will stop working.

Windows quick check

OK	The browser displays Wind Turbine Design Lab.
OK	The Design Lab, Guide, and Glossary tabs are visible.
OK	Changing an input updates the estimated results.
OK	No Deploy or payment step is required.

MACOS - PART 1

Install Git and uv

Complete these steps once using the Terminal application.

1 Open Terminal

Press **Command + Space**, type **Terminal**, and press Enter.

2 Check Git

Run:

```
git --version
```

If macOS asks to install developer tools

Choose **Install** and wait for Apple Command Line Tools to finish. If no prompt appears and Git is missing, run the command below.

```
xcode-select --install
```

3 Install uv

Copy and run:

```
curl -Lsf https://astral.sh/uv/install.sh | sh
```

4 Restart Terminal and confirm uv

Close Terminal, open it again, and run:

```
uv --version
```

5 Go to Documents

Run:

```
cd ~/Documents
```

6 Download the project

Run:

```
git clone https://github.com/TBK0749/wind-turbine-design-lab.git
```

7 Enter the folder and install packages

Run these commands one at a time:

```
cd wind-turbine-design-lab
```

```
uv sync
```

Be patient on the first run

uv may download Python and all project dependencies. Wait until the Terminal prompt returns.

MACOS - PART 2

Start and stop the simulator

Use this shorter process after the first installation.

Start the app

1 Open Terminal

Press Command + Space, search for Terminal, and open it.

2 Go to the project folder

Run:

```
cd ~/Documents/wind-turbine-design-lab
```

3 Start Streamlit

Run:

```
uv run streamlit run app/main.py
```

4 Open the local address

If the browser does not open automatically, enter:

```
http://127.0.0.1:8501
```

Stop the app

Return to the Terminal window and press **Control + C**. On Mac this is the Control key, not the Command key.

What the browser address means

127.0.0.1 means this computer. 8501 is the local port used by Streamlit. The page is not public on the internet and other people cannot use it unless they install and run their own copy.

Safe to close?

Save or download any result files you need, stop Streamlit with Control + C, and then close Terminal.

FIRST USE

A simple tour of the simulator

Installation is complete when the dashboard opens in your browser.

Area	What to do
Design inputs	Enter wind speed, blade length, number of blades, generator settings, and trial time.
Blade geometry table	Enter each measured blade station: radial position, chord, and twist.
Airfoil	Choose the option that best matches the blade cross-section you will build.
Results	Review competition power in mW, voltage, current, RPM, Cp, TSR, and warnings.
Guide	Read the competition workflow and installation/update notes.
Glossary	Look up engineering terms such as chord, twist, Cp, TSR, stall, and BEMT.

Recommended first test

O K	Leave the default 50 cm blade preset selected.
O K	Confirm the section table contains position, chord, and twist values.
O K	Change wind speed slightly and watch the competition mW result.
O K	Open the Guide and Glossary tabs.
O K	Download one CSV or JSON result to confirm browser downloads work.

Educational model

The simulator compares design ideas and explains trade-offs. The measured competition result is still the final judge.

UPDATES

Get the latest version

You do not need to clone the project again if you installed it with Git.

Windows

Open PowerShell and run each line:

```
cd $HOME\Documents\wind-turbine-design-lab

git pull

uv sync

uv run streamlit run app/main.py
```

macOS

Open Terminal and run each line:

```
cd ~/Documents/wind-turbine-design-lab

git pull

uv sync

uv run streamlit run app/main.py
```

What the update commands mean

Command	Purpose
<code>git pull</code>	Downloads new project files from GitHub.
<code>uv sync</code>	Updates the Python environment if project requirements changed.
<code>uv run ...</code>	Starts the latest installed version.

Protect your own files

Keep exported CSV, JSON, images, and experiment notes outside the project folder. Avoid editing project source files, because local edits can block `git pull`.

If you used Download ZIP

ZIP installations cannot use `git pull`. Download a new ZIP, unzip it, and replace the old folder. Installing with Git is recommended for easier updates.

HELP - PART 1

Common installation problems

Most problems are caused by being in the wrong folder or by a terminal that needs restarting.

git is not recognized

Likely cause

Git is not installed, or PowerShell was open during installation.

Try this: Install Git, close every PowerShell window, reopen it, and run `git --version`.

uv is not recognized

Likely cause

The uv installer completed but the current terminal has not refreshed its PATH.

Try this: Close and reopen PowerShell or Terminal. If it still fails, run the uv installation command again.

No such file or directory / path not found

Likely cause

The terminal is not in the folder expected by the command.

Try this: Use the exact `cd` command shown for your operating system.

destination path already exists

Likely cause

A folder named `wind-turbine-design-lab` is already present.

Try this: Do not clone again. Enter the existing folder and use `git pull`.

Failed to connect / download

Likely cause

The computer is offline, a proxy is blocking downloads, or GitHub is temporarily unavailable.

Try this: Check the internet connection and try again. School or company networks may require IT assistance.

HELP - PART 2

Problems while starting the app

Read the last lines in the terminal first - they usually identify the issue.

The browser did not open

Keep the terminal running. Open your browser manually and visit `http://127.0.0.1:8501`.

This site cannot be reached

The Streamlit command is not currently running, or it stopped with an error. Return to the terminal, enter the project folder, and run the startup command again.

Port 8501 is already in use

Another copy of Streamlit is probably still running. Find its PowerShell or Terminal window and press **Ctrl + C** (Windows) or **Control + C** (Mac). Then start again.

Streamlit asks for an email

Leave the field blank and press Enter. Registration is optional.

git pull says local changes would be overwritten

Someone changed files inside the project folder. Copy any important personal files out of the folder before asking the project maintainer for help. Do not delete work you need.

The dashboard displays a red Python error

O K	Copy or photograph the complete red error message.
O K	Return to the terminal and copy the last 20 lines.
O K	Run <code>git pull</code> , then <code>uv sync</code> .
O K	Restart the app. If the error remains, send the captured information to the maintainer.

Do not share secrets

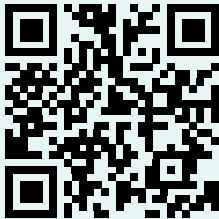
Screenshots should not include passwords, access tokens, private school data, or unrelated personal files.

QUICK REFERENCE

Commands worth saving

Print this page or keep it near the competition workstation.

Task	Windows PowerShell	macOS Terminal
Open project folder	<code>cd \$HOME\Documents\ wind-turbine-design-lab</code>	<code>cd ~/Documents/ wind-turbine-design-lab</code>
Install / repair	<code>uv sync</code>	<code>uv sync</code>
Start app	<code>uv run streamlit run app/main.py</code>	<code>uv run streamlit run app/main.py</code>
Update app	<code>git pull uv sync</code>	<code>git pull uv sync</code>
Stop app	Ctrl + C	Control + C
Open in browser	<code>http://127.0.0.1:8501</code>	<code>http://127.0.0.1:8501</code>



Project repository

<https://github.com/TBK0749/wind-turbine-design-lab>

Scan the QR code or open the link to view the source code, README, and latest updates.

REFERENCE

Official links and notes

Use these sources if an installer screen changes after this guide was published.

Source	Link and purpose
Project repository	https://github.com/TBK0749/wind-turbine-design-lab The latest project code, README, and release history.
Installing Git on Windows	https://git-scm.com/install/windows Official Git download page for Windows.
Installing Git on macOS	https://git-scm.com/install/mac Official Git installation options for macOS.
Installing uv	https://docs.astral.sh/uv/getting-started/installation/ Official uv installation commands for Windows, macOS, and Linux.
Python management with uv	https://docs.astral.sh/uv/guides/install-python/ Explains how uv automatically downloads a required Python version.
Cloning a repository	https://docs.github.com/en/repositories/creating-and-managing-repositories/cloning-a-repository GitHub's official explanation of cloning a repository.

Scope

This guide covers local installation for Windows and macOS. It does not require Streamlit Community Cloud, deployment, or a paid hosting service.

Installation complete?

You are ready when the dashboard opens at `http://127.0.0.1:8501` and responds when an input changes.